Segment Routing
–もうLDP/RSVPはいらない–

Shishio Tsuchiya

shtsuchi@cisco.com

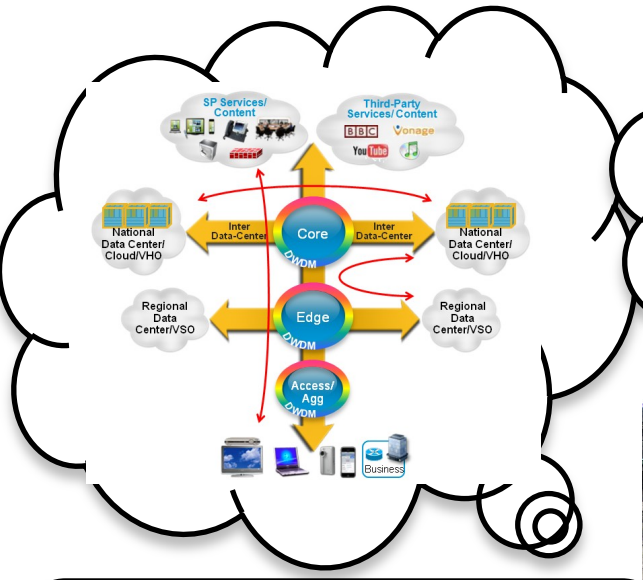
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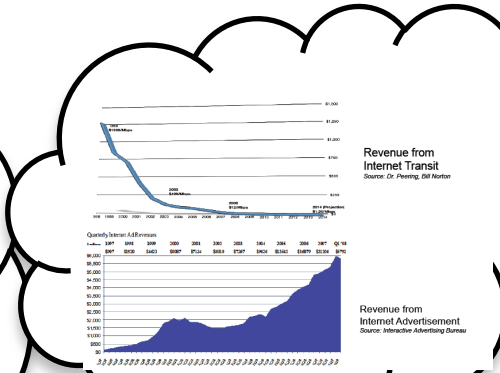
Agenda

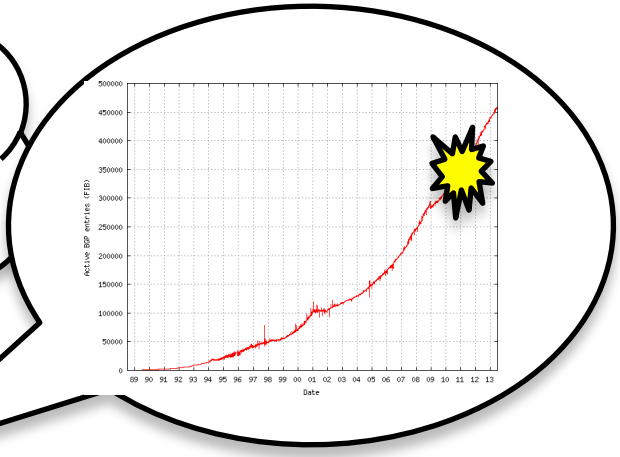
- Segment Routing Overview
- Complex of legacy MPLS
- Segment Routing use case
- IETF Status

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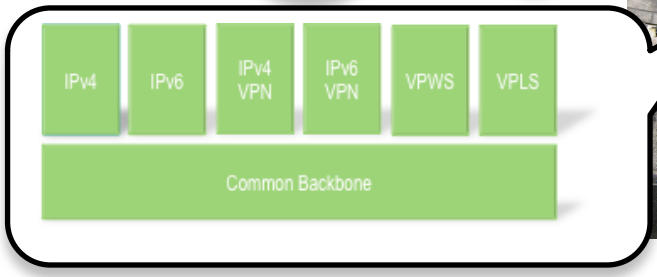
サービス・プロバイダーの憂鬱

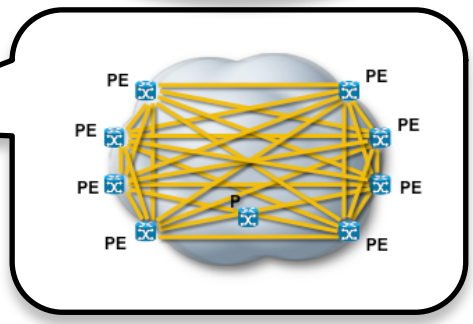












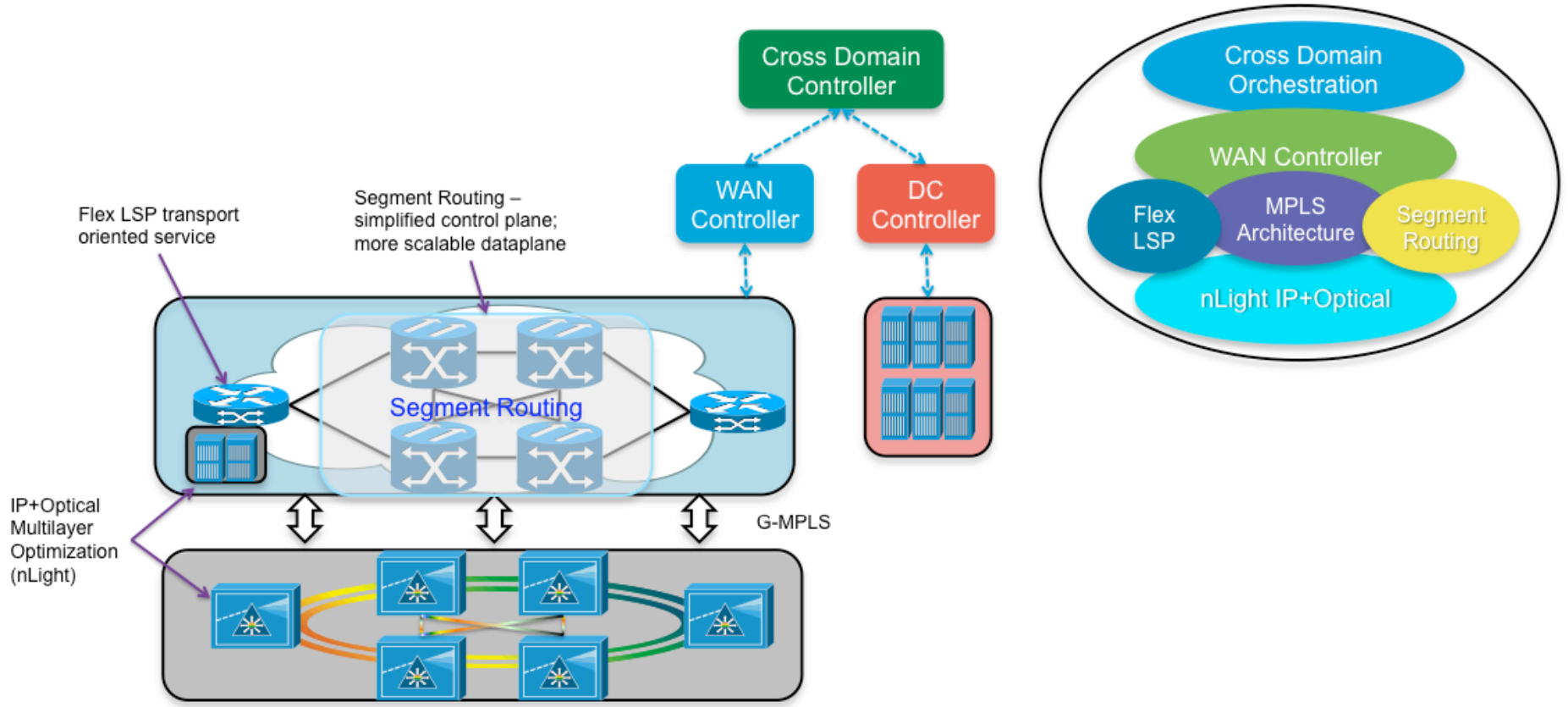
- トラフィック・パターンの多様化
- レベニューモデルの変化
- 経路爆発

- マルチサービス/共通基盤
- パス・スケーラビリティ
- 管理コスト

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MPLSDN

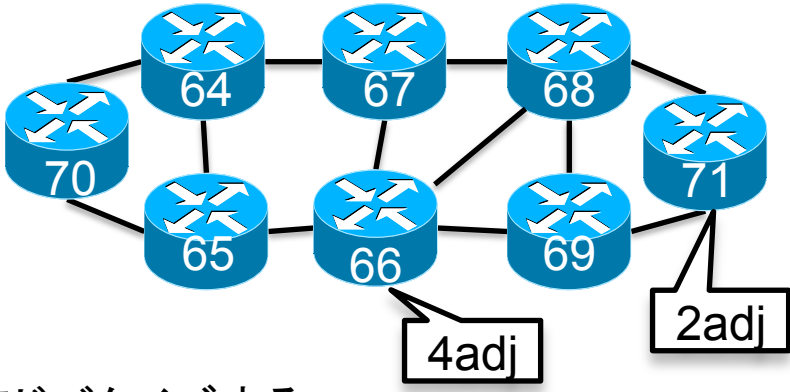
管理コストと複雑さを増加させずに、より速くサービスを達成する方法とは？



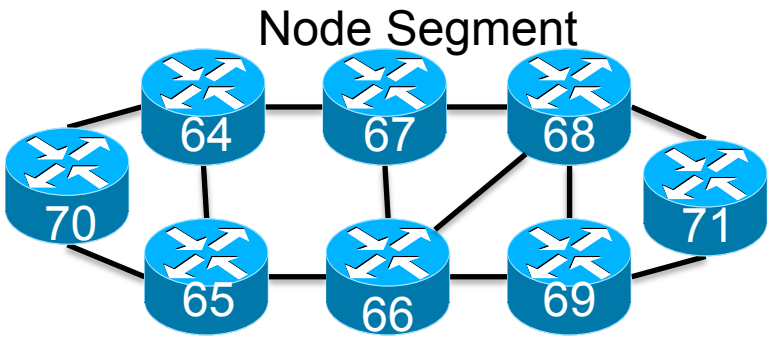
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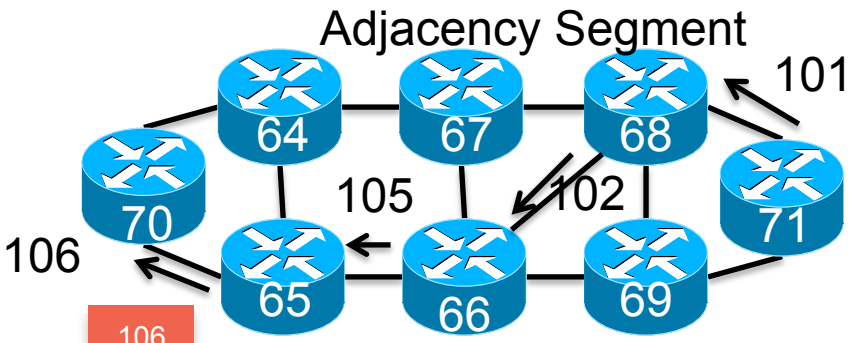
Segment Routing

draft-previdi-filsfils-isis-segment-routing



- ネットワークを*Segment*で表現する
- SegmentにはNodeとAdjacency2つの要素がある
- LDP/RSVPを使わず、直接IGPによりこれらのIDをアドバタイズする
- ネットワークからLDP/RSVPのステートを排除する事が出来る
- 現在はdraft-filsfils-rtgwg-segment-routingにてアーキテクチャーを定義



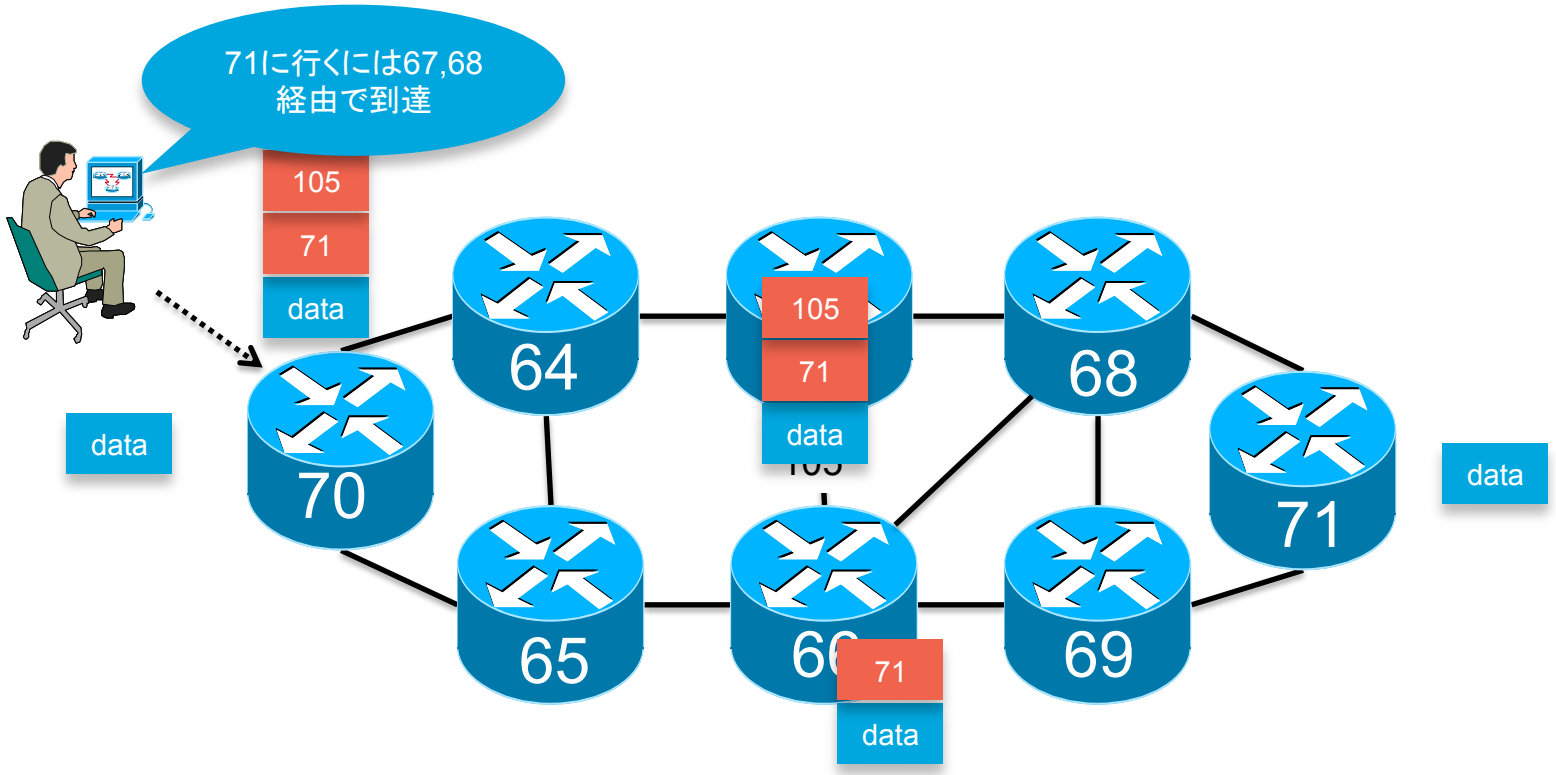




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Segment Routing
Combination NodeID+AdjID

draft-previdi-filsfils-isis-segment-routing



- Node IDとAdjacency IDの組み合わせも可能

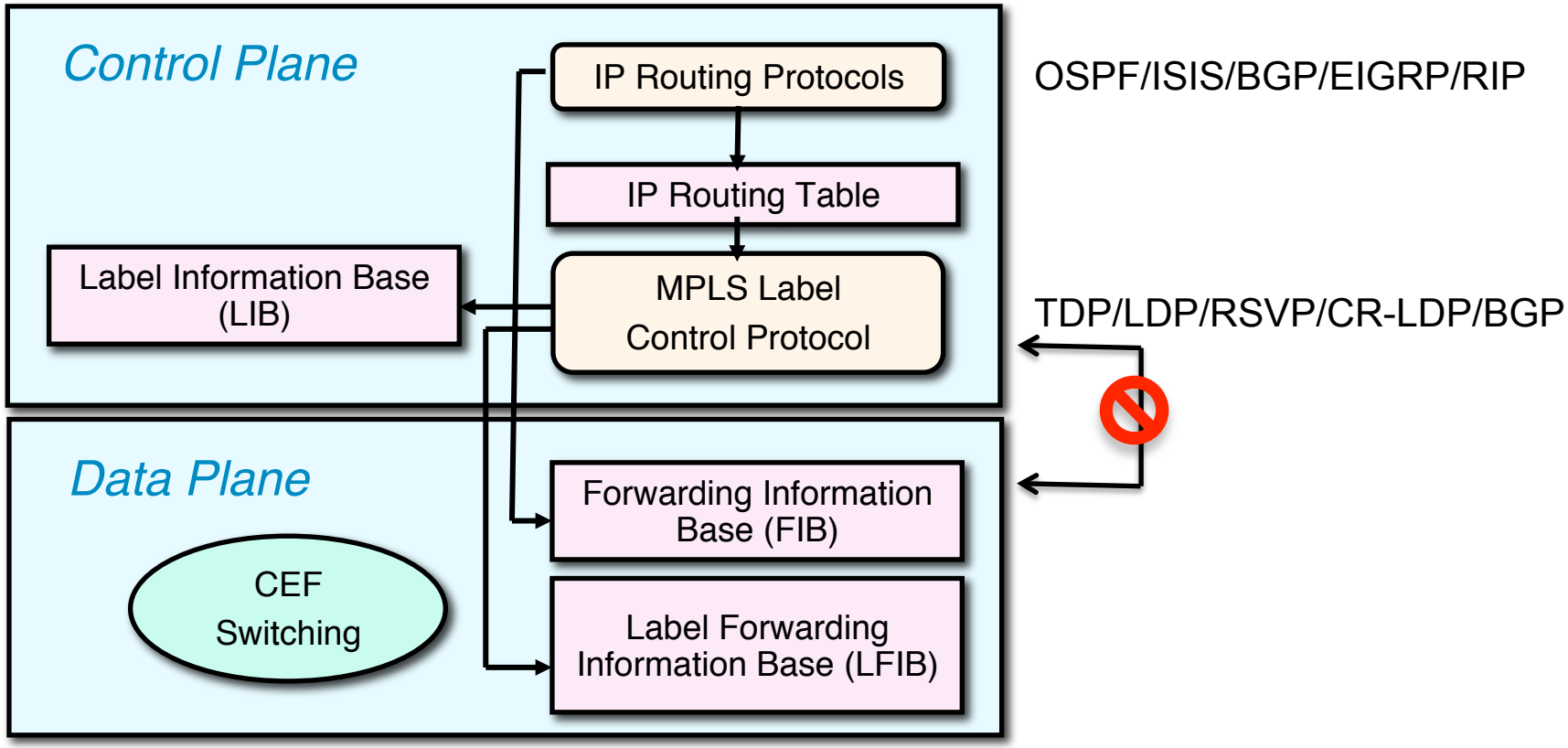
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Agenda

- Segment Routing Overview
- Complex of legacy MPLS
- Segment Routing use case
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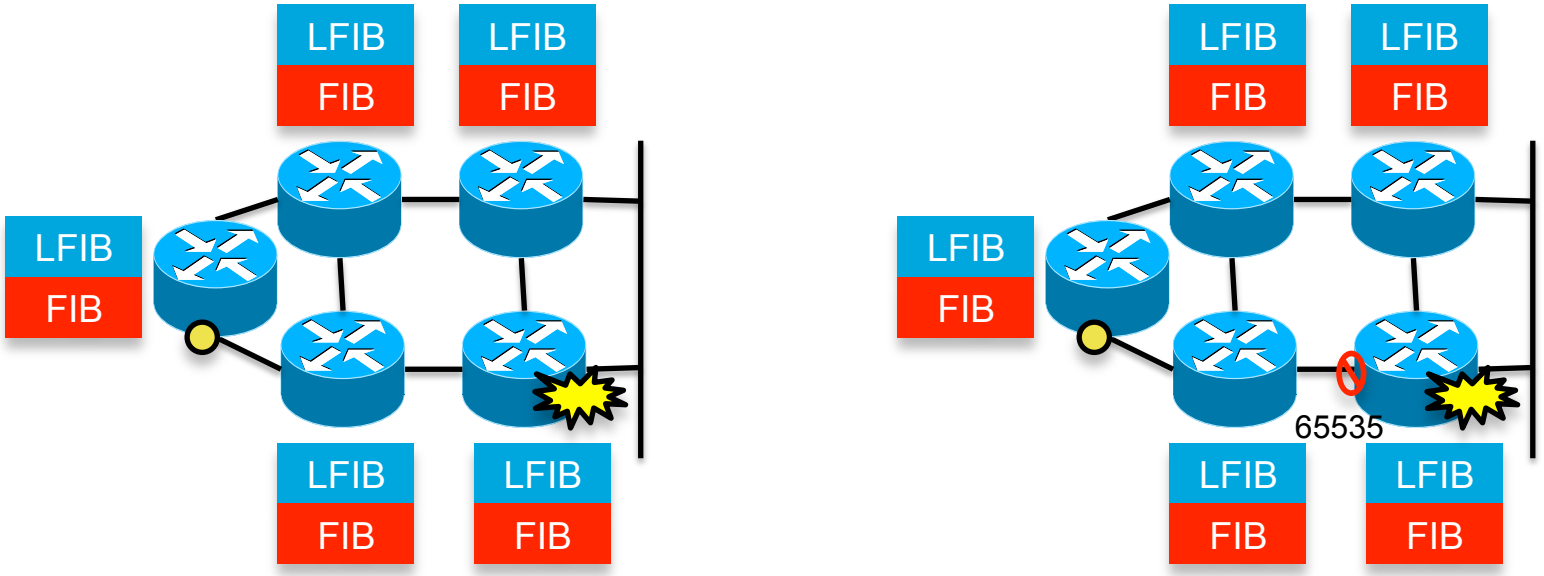
MPLS OAM Requirement



- コントロールプレーンとデータプレーンを分離
- データプレーンを確認するための新しいOAMツール(RFC4379)が必要に

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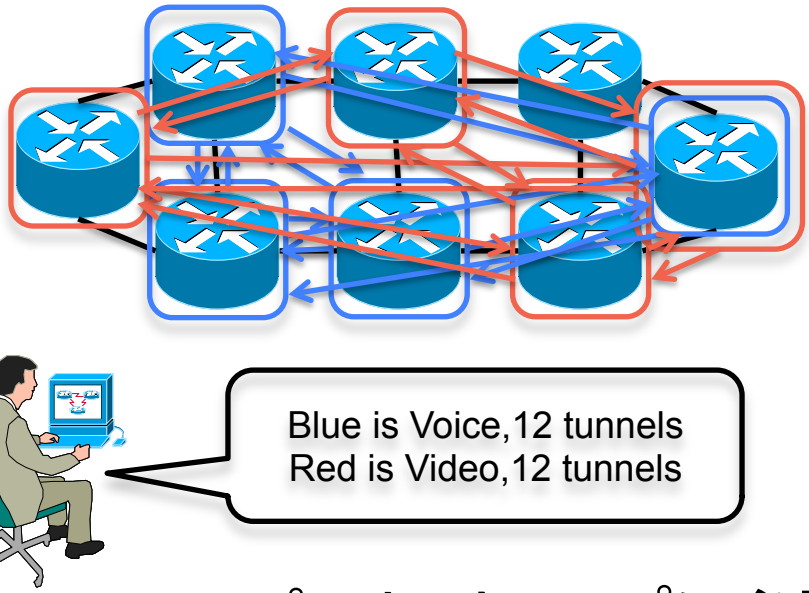
MPLS LDP IGP Synchronization(RFC5443)



- 2つ以上のコントロールプレーン/データプレーンを使う為に相互での矛盾が起こりうる
- LDP Stateの監視を行いMax Metricでコントロールをする

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Discovery of MPLS LSR TE Mesh Membership (RFC4972)



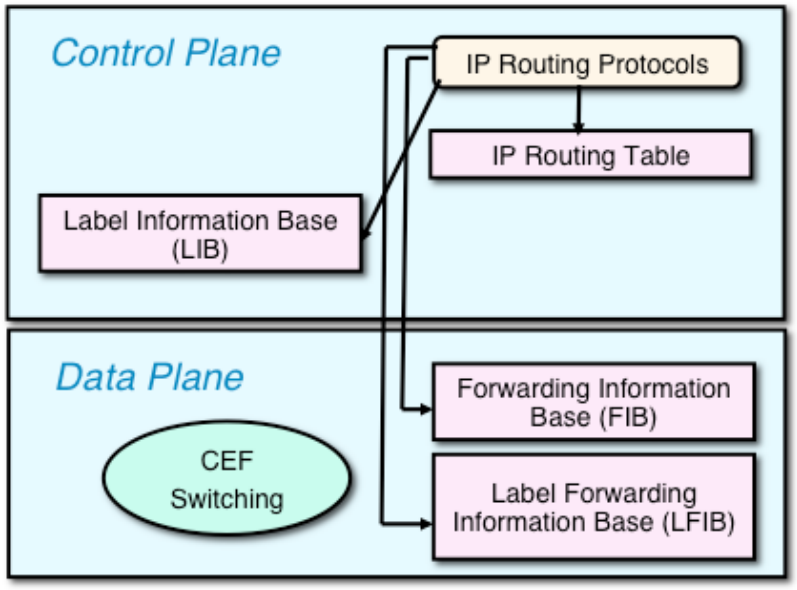

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      mesh-group-number 1                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                     Tail-end IPv4 address 1                   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Name length  |               Tail-end name 1                 |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
//                                                               //
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      mesh-group-number n                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                     Tail-end IPv4 address n                   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Name length  |               Tail-end name n                 |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

- RSVPのパスセットアップに必要なTEDBの為にOSPF/ISISを拡張(RFC3630/RFC5305)
- ノードがどのMesh Groupに入るかの伝播も拡張し、設定負荷などを軽減

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まとめ

- コントロールプレーンとデータプレーンが大きく離れているので、追加でOAMが必要に
- コントロールプレーンが複数あるので、監視する仕組みが必要に
- 多くの設定を要する為に、設定情報を分散させる仕組みが必要に



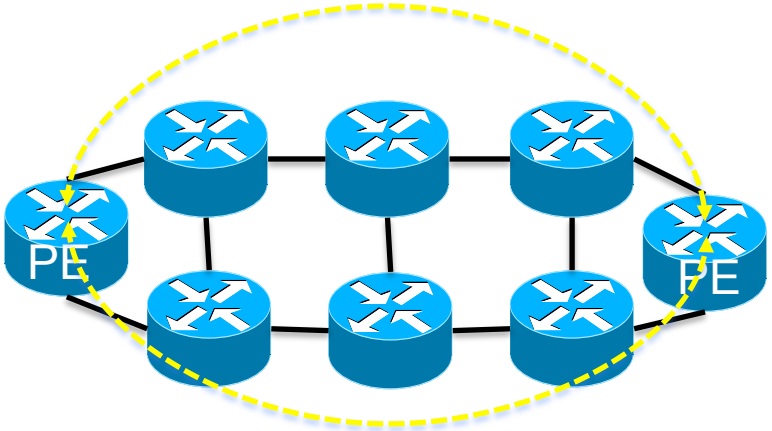
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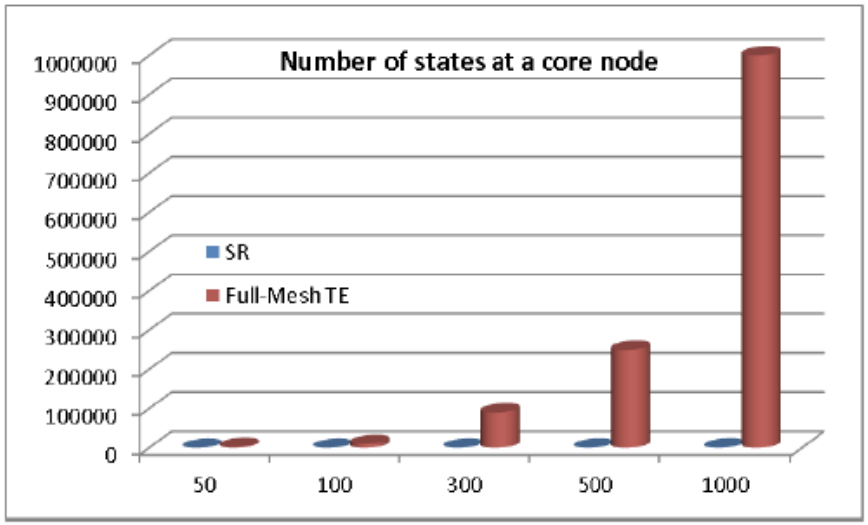
Agenda

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Applicability
Simplicity/Scalable TE

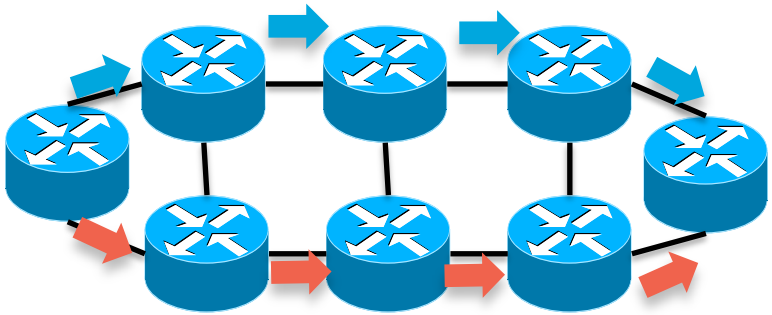


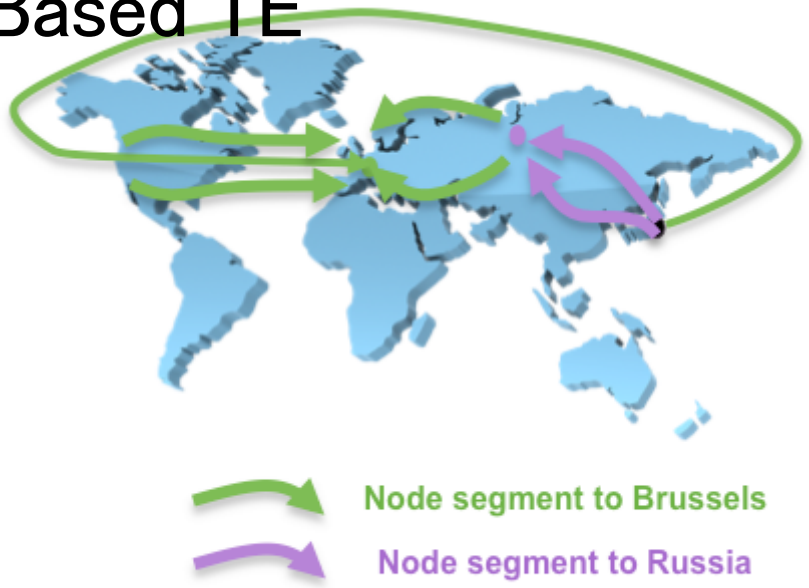


- Simplicity
  VPNサービスの為にはPEのNode IDを知ってるのみで良い
  IGP/LDPのSyncなどを気にする必要が無くなる
  一つのプロトコルのみの運用
- Scalable TE
  RSVP TEでフルメッシュの運用をすると、N-1のHead/TailおよびN(N-1)/2 midpoint LSPの管理が必要
  SRではStateを管理する必要が無い。State in the packet

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Applicability
Simple Disjointness/CoS Based TE

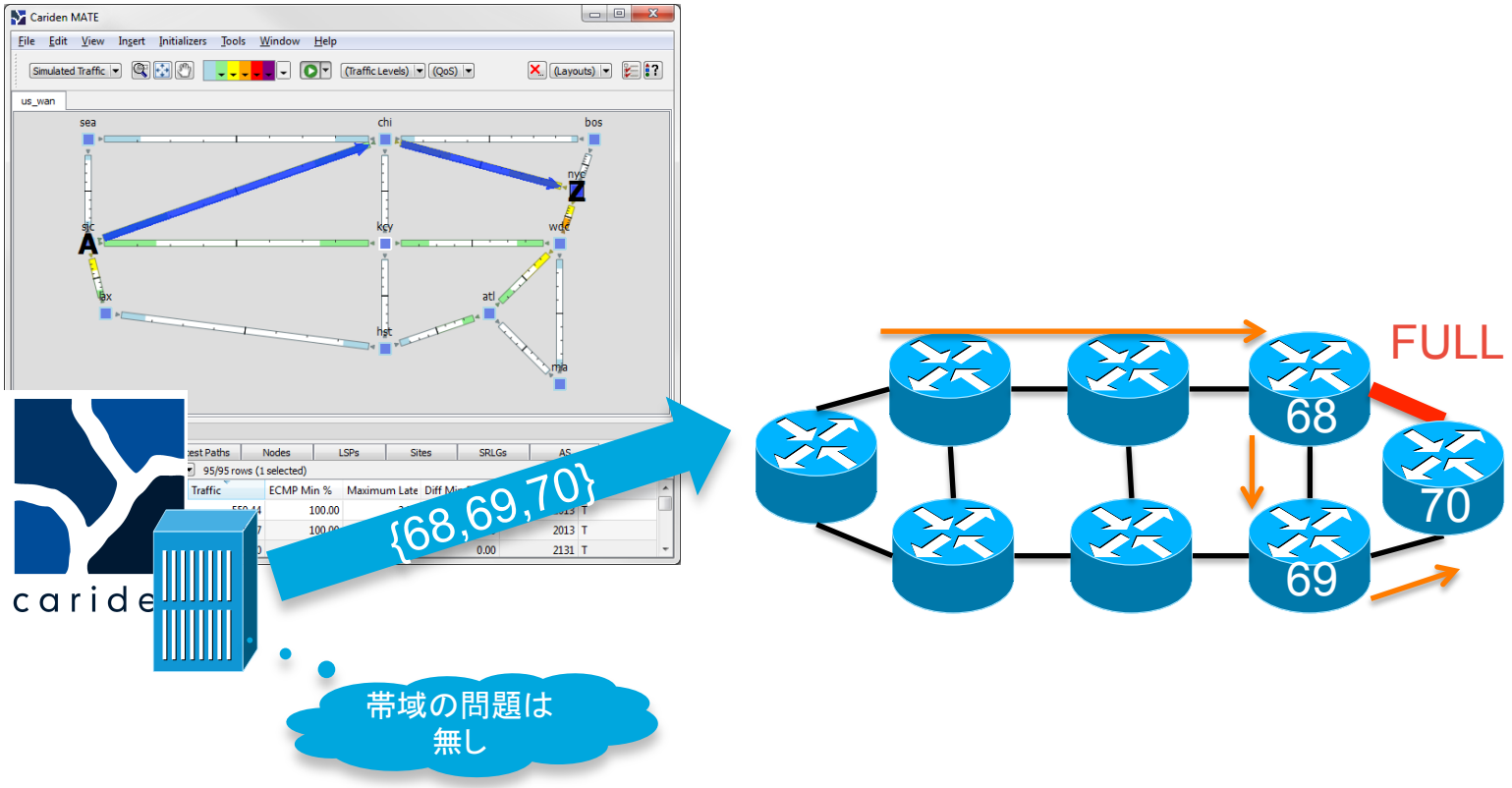




- Simple Disjointness
  品質の厳しい顧客にはプライマリ/セカンダリが互いに分離したパスが必要になる。
  SRであればNode IDをStackする事でコントロールが可能
- CoS Based TE
  データは回線コストの安いUS経由/VoIPは低遅延のロシア経由など
  品質に応じたトラフィックエンジニアリングも実施可能

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Applicability
Segment Routing in Software Defined Networks (SR-SDN)



- ネットワークがシンプルなので、パスセットアップも非常に簡単になる

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Agenda

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Segment Routing IETF Status

- IETF 87 – Berlin, Germany

  Stacked Tunnels for Source Routing (STATUS) BOF

  http://www.ietf.org/proceedings/87/status.html

- IETF 88 – Vancouver, BC, Canada

  "Source Packet Routing in Networking" (spring) WG

  https://datatracker.ietf.org/doc/charter-ietf-spring/

  Milestones:

  - ✓ High Level Architecture
  - ✓ MPLS Architecture/IPv6 Architecture
  - ✓ Use case
  - ✓ OAM/Inter-work/interop report etc..

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IETF Activity

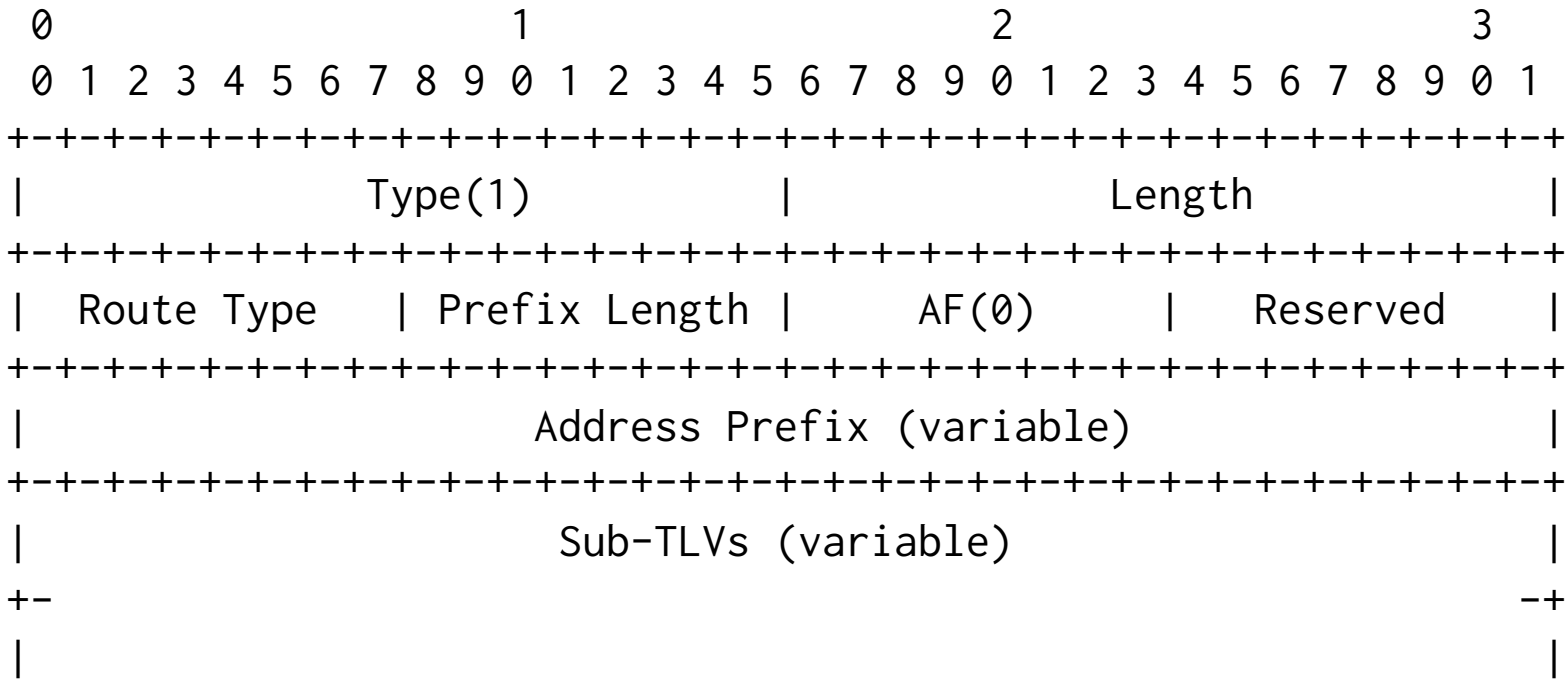
http://tools.ietf.org/wg/spring/

- SRアークテクチャー
  draft-filsfils-rtgwg-segment-routing
- Use case
  draft-filsfils-rtgwg-segment-routing-use-cases
- SRの為のISIS拡張
  draft-previdi-isis-segment-routing-extensions
- SRの為のOSPF拡張
  draft-psenak-ospf-segment-routing-extensions
  draft-psenak-ospf-segment-routing-ospfv3-extension
- SR with MPLS
  draft-filsfils-spring-segment-routing-mpls
  draft-filsfils-spring-segment-routing-ldp-interop
  draft-kumar-mpls-spring-lsp-ping

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OSPF Extended Prefix TLV

draft-psenak-ospf-segment-routing-extensions

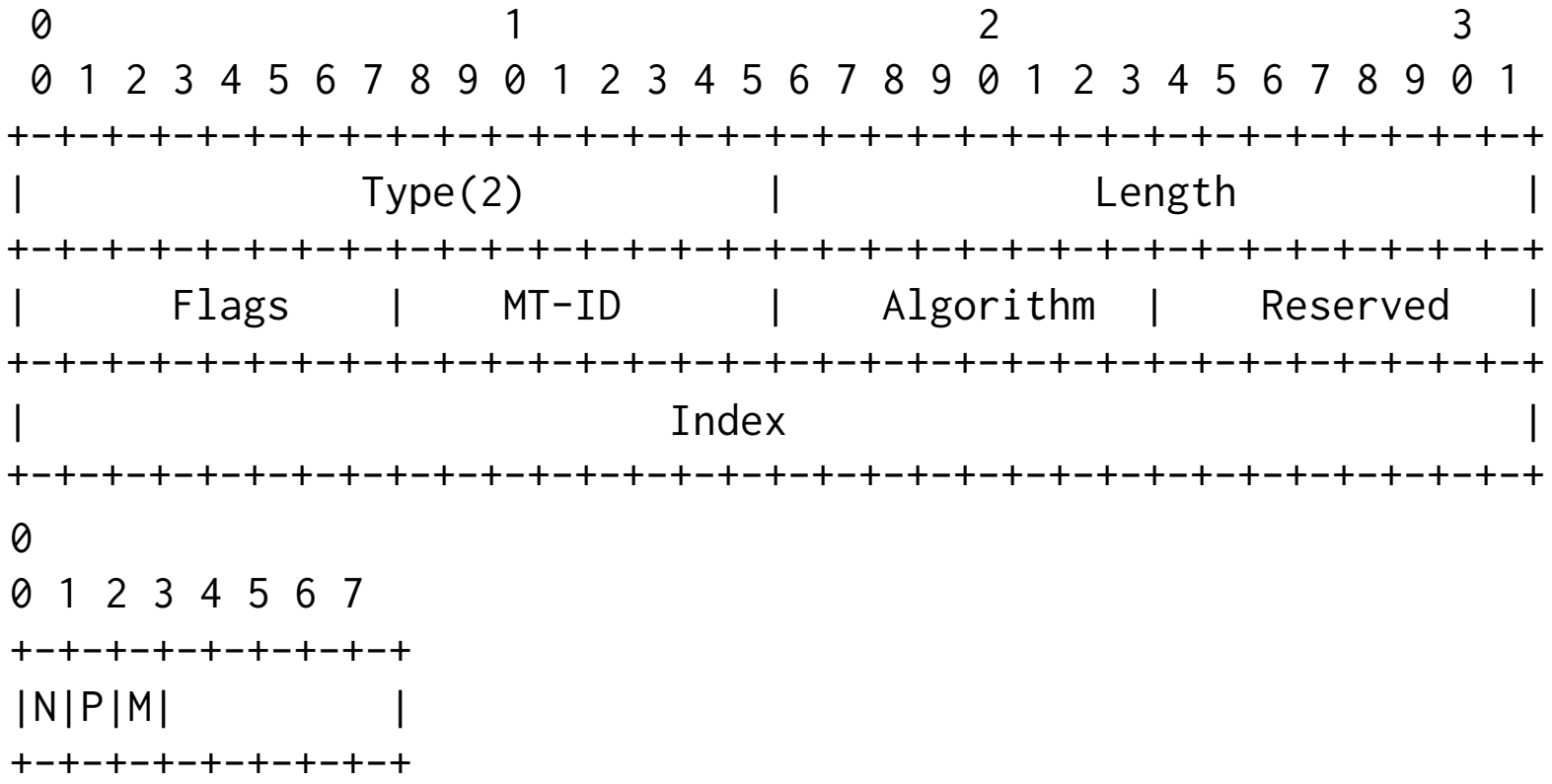


OSPFルートタイプ:
0 - unspecified
1 - intra-area
3 - inter-area
5 - external
7 - NSSA external
AF:0 IPv4 unicast

- Opaque TLVを用いてSIDを伝播

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Prefix SID Sub-TLV draft-psenak-ospf-segment-routing-extensions

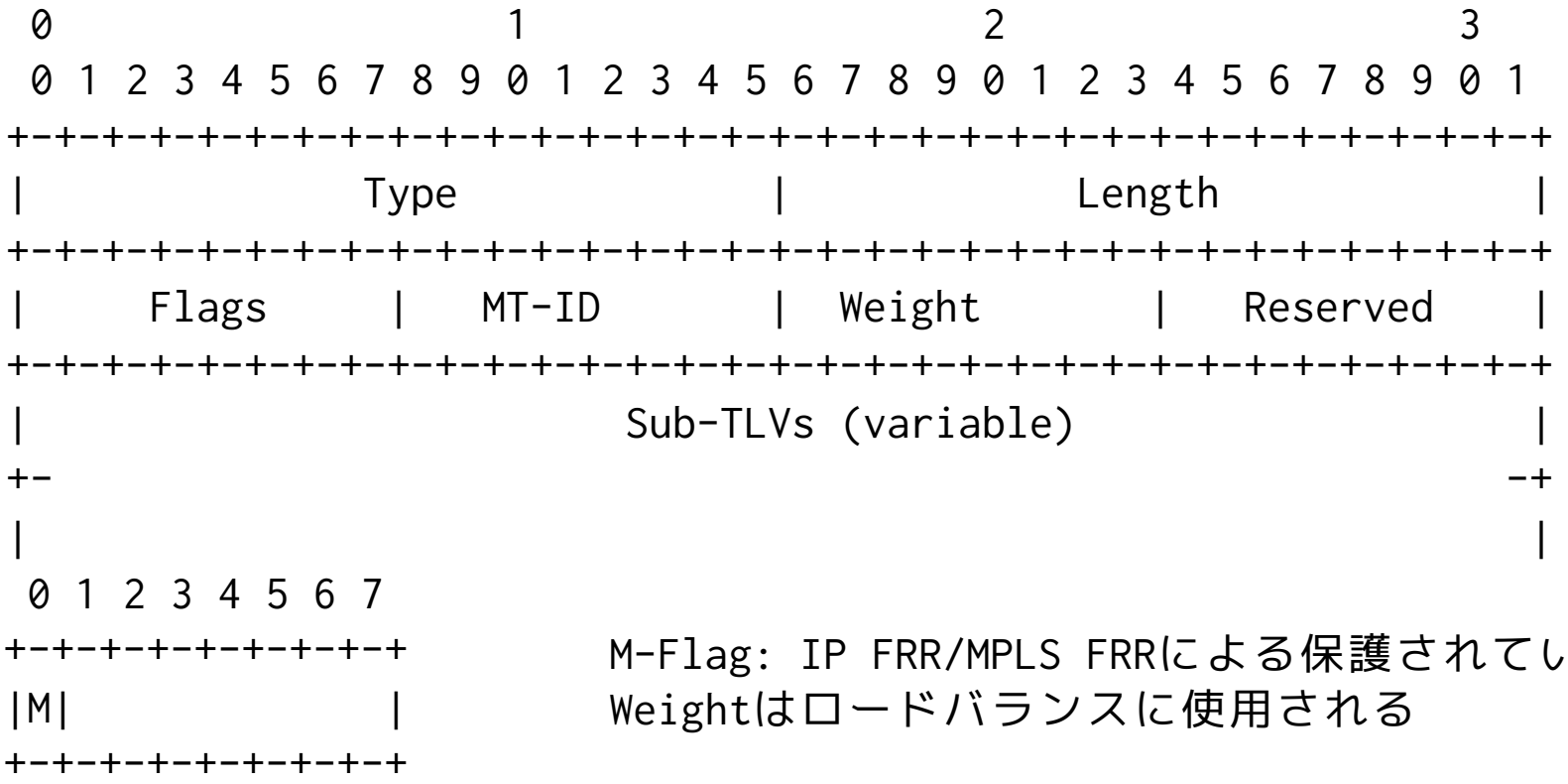


- N-Flag: Node-SID flag.NodeSIDかどうかを示す。
- P-Flag: no-PHP flag.セットしてあればPHPをしてはならない
- M-Flag: Mapping Server Flag.セットしてあればMapping Serverによりアドバタイズされた

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SID/Label Binding sub-TLV

draft-psenak-ospf-segment-routing-extensions

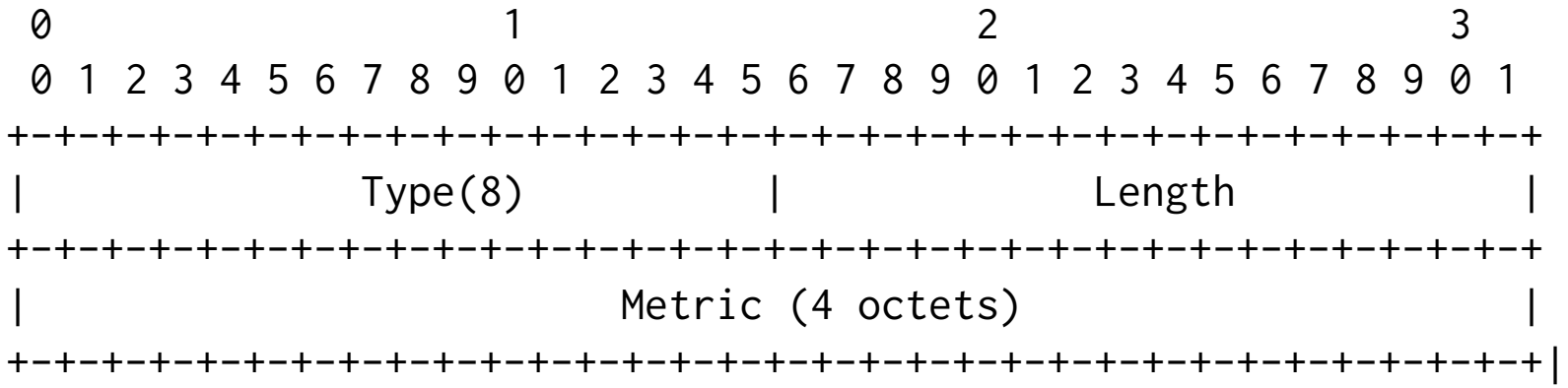


- Prefixへのパスを示す
- SubTLVにはERO Metric sub-TLV/IPv4 ERO sub-TLV

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ERO Metric sub-TLV

draft-psenak-ospf-segment-routing-extensions



- Metric

  Source/Destinationのコスト比較の為に使用する

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IPv4 ERO subTLV

draft-psenak-ospf-segment-routing-extensions



- Backup ERO/unnumbered ERO subTLVなどがある

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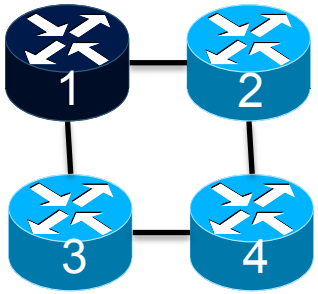
SR Prototype

- Images available for ASR9000, CRS and XR12000
- Prototype supports
  - MPLS data plane
  - Node segment id
  - IS-IS
- Implementation based on 4.3.1 code
- Available for customer lab use
- CLI not final

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Nodal–id configuration

```
router isis abc
 net 47.0000.0000.0000.1111.00
 address-family ipv4 unicast
  metric-style wide
 !
 address-family ipv6 unicast
 !
 interface Loopback0
  passive
  address-family ipv4 unicast
   nodal-sid sid-value 16001
  !
  address-family ipv6 unicast
   nodal-sid sid-value 16901
  !
```



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show isis database

```
RP/0/RSP0/CPU0:ASR9k-1#sh isis database verbose
IS-IS abc (Level-1) Link State Database

LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime  ATT/P/OL

ASR9k-1.00-00       * 0x00000057   0x9e4e        675             0/0/0

  Area Address: 47.0000

  NLPID:        0xcc

  NLPID:        0x8e

  MT:           Standard (IPv4 Unicast)

  MT:           IPv6 Unicast                                     0/0/0

  Hostname:     ASR9k-1

  IP Address:   10.1.255.1

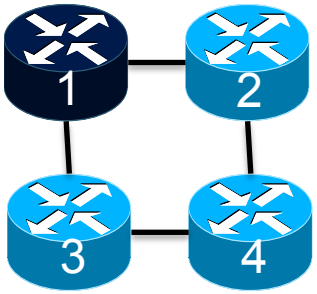
  IPv6 Address: 2001:db8:ffff::1

  Metric: 10         IS-Extended ASR9k-2.00

  Metric: 10         IP-Extended 10.1.13.0/24

  Metric: 0          IP-Extended 10.1.255.1/32

    Nodal-SID: 16001 PHP-off:0 Ext:0
```

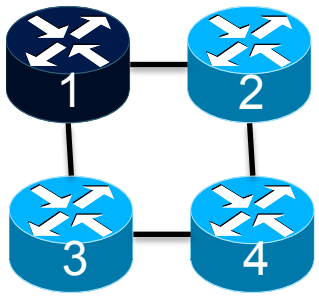


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show mpls fowarding

```
RP/0/RSP0/CPU0:ASR9k-1#sh mpls for

Local  Outgoing    Prefix            Outgoing     Next Hop        Bytes
Label  Label       or ID             Interface                    Switched
------ ----------- ----------------- ------------ --------------- ------------
16001Aggregate   default: Per-VRF Aggr[V]   \   default           0           0
16002  Pop         No ID             Te0/0/0/0    10.1.12.2       752
16003  Pop         No ID             Te0/0/0/1    10.1.13.3       0
16004  16004       No ID             Te0/0/0/0    10.1.12.2       0
       16004       No ID             Te0/0/0/1    10.1.13.3       0
16902  Pop         No ID             Te0/0/0/0    fe80::6e9c:edff:fe30:de0   \ 1112
16903  Pop         No ID             Te0/0/0/1    fe80::4255:39ff:fe7f:fbe9   \ 0
16904  16904       No ID             Te0/0/0/0    fe80::6e9c:edff:fe30:de0   \ 0
       16904       No ID             Te0/0/0/1    fe80::4255:39ff:fe7f:fbe9   \ 0
```



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show cef x.x.x.x

```
RP/0/RSP0/CPU0:ASR9k-1#sh cef 10.1.255.4

Wed Jun 26 15:34:16.877 UTC

10.1.255.4/32, version 10515396, internal 0x4004001 (ptr 0x7630ff20) [1], 0x0 (0x71202dd8), 0x450 (0x7665c050)

 Updated Jun 26 15:26:10.750

 remote adjacency to TenGigE0/0/0/0

 Prefix Len 32, traffic index 0, precedence n/a, priority 1

   via 10.1.12.2, TenGigE0/0/0/0, 8 dependencies, weight 0, class 0 [flags 0x0]

    path-idx 0 [0x72372224 0x0]

    next hop 10.1.12.2

    remote adjacency

     local label 16004      labels imposed {16004}

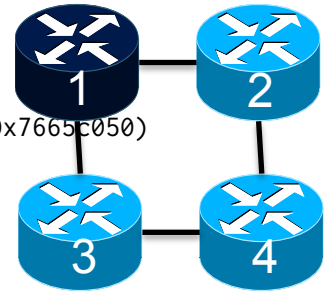
   via 10.1.13.3, TenGigE0/0/0/1, 9 dependencies, weight 0, class 0 [flags 0x0]

    path-idx 1 [0x723721c8 0x0]

    next hop 10.1.13.3

    remote adjacency

     local label 16004      labels imposed {16004}
```



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まとめ

- Segment RoutingではOSPF/ISISにて直接データプレーン情報をアドバタイズする事で複雑さを回避/ステータスの管理を軽減する
- シンプルにする事により、アプリケーションとの連動をしやすいコアネットワークを構築可能に

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Thank you.

